

Daniel Hyman

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Education

University of Guelph

Guelph, ON

Bachelor of Engineering: Engineering Systems and Computing (Co-op) Expected Graduation: April 2029

Relevant Courses: Engineering Design, Data Structures, Digital Hardware, Electric Circuits, Kinematics

Certification: CSWA SolidWorks Associate

Technical Skills

Programming: Python, C/C++, MATLAB, VHDL, SQL, Java, JavaScript

Frameworks & Libraries: ROS2, React, Django, FastAPI, RestAPI, Node.js

CAD & Design: SolidWorks, AutoCAD, KiCAD

Hardware & Embedded: ESP32, STM32, FPGA, Vivado

Developer Tools: Linux, Git, Docker, PostgreSQL, PgAdmin4

Work Experience

Full Stack Developer | Remote | Zoltaro Inc.

May - Aug 2026

- Developed and documented a **PostgreSQL** database workflow using PgAdmin4, mapping database components and their integration with a Python-based backend
- Developed AI training content for the company's web platform and configured virtual environments to test the frontend and backend systems
- Researched **AI**, **ML**, LMS platforms, and MLOps to identify existing market products and customer demand regarding AI prompting tools and support resources

Electrical Engineer | Guelph, ON | Rocketry and Robotics Clubs

Jan - Jun 2026

- Collaborating with the avionics team to design and launch a solar-powered **CubeSat Satellite** into the lower atmosphere by August 2026
- Developed a control board prototype implementing an STM32 Blackpill using KiCAD to read pressure from 6 Op Amps, and sends signals to 4 Solenoids, extending to exterior functionalities such as propulsion

Projects

Autonomous Robotic Arm | ROS2, C++, Python, YOLO, SolidWorks

May 2026 - Present

- Designed and 3D-printed a multi-jointed robotic arm in **SolidWorks**, integrating an ESP32, servo motors, and potentiometers for embedded control
- Developed an **AI vision** system using YOLO and Python to detect objects by color and determine their position within the camera frame, enabling autonomous sorting
- Integrated the robotic arm, ESP32 camera, and vision system using **ROS2**; created URDF and RViz2 configurations to model and visualize the arm's kinematics and joint movements

Leaf Watch | Python, React, FastAPI, Node.js, NASA API

Nov 2025

- Uses **NASA satellite data** to monitor deforestation rates and an **ML model** to predict future trends
- Implemented the map UI and timelapse feature to visualize deforestation, data analytics through the **NASA API**, and ML model training

Four-bit Arithmetic Logic Unit | VHDL, Vivado, FPGA

Nov 2025

- Developed an ALU capable of 8 arithmetic and 4 logic operations.
- Designed hierarchical digital logic from half-adders to a 4-bit ALU and 7-segment **FPGA** display
- Simulated all input combinations and optimized FSM and logic equations